

Jessica Chou

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OBJECTIVE

Seeking a Sr. Software Engineering role to leverage extensive expertise in big data and cloud technologies, with a focus on architecting scalable, high-performance systems in fast-paced environments.

TECHNICAL SKILLS

Big Data Technologies:	Flink, Spark, Kafka, Zeppelin, Delta Lake, Hadoop, S3, AWS EMR, AWS EKS,
Enterprise Web Technologies:	REST, SOAP, Servlet, Tomcat, Lucene/Solr4, Hibernate
Web:	Nginx, Apache2, PHP5, custom PHP extensions, HTML, CSS, JavaScript
MVC Framework:	Spring Boot, Struts2, Zend Framework, CakePHP
Scalability:	Apache Kafka, ActiveMQ, RabbitMQ, Apache Camel
Cloud Technologies:	Azure, AWS EC2, S3, RDS Oracle/MySQL, Redshift
Operating Systems:	Linux (Ubuntu, Redhat), Windows, UNIX (Solaris, AIX, Linux, HP-UX)
Languages:	Java, Scala, Python, PHP 5, JavaScript, Perl, C++, C, Visual C++, shell, SQL
Databases:	Postgres, Progress, Oracle, MySQL, MongoDB, CosmosDB, Redis, Memcache, JDBC

PROFESSIONAL EXPERIENCE

9/2020 – 10/2024, Samsung Research America

Sr. Staff Engineer

- Architected, prototyped, developed, and deployed data pipelines for Samsung Ads advertising platform using Apache Flink, Spark, Kafka, Delta Lake S3, Scala, Java, Zeppelin, AWS Athena, Docker, Kubernetes, Prometheus, AWS EMR, AWS EKS, Grafana, Kibana
- Implemented real-time streaming pipeline job processing raw TV data enabling instant data insights utilizing Apache Flink, Kafka, Delta Lake S3, and Scala at about 1 petabyte data daily
- Implemented batch processing pipeline job processing raw TV data for backfill purposes with Apache Spark, Delta Lake on S3, AWS EMR and Scala
- Ensured high availability and observability with AWS EKS, Docker, Prometheus, Grafana, and Kibana
- Implemented tools with Zeppelin notebook and AWS Athena to do data quality, data analysis, and troubleshooting
- Performance tuning, profiling, troubleshooting, and configuration of Apache Flink jobs and Apache Spark jobs
- PoC Generative AI app on TV using Apache Tomcat, Spring Boot, Kafka, Redis, Python, Neo4J Graph database

10/2016 – 9/2020, Adobe

Sr. Computer Scientist

Adobe Campaign on Adobe Experience Platform and Adobe Creative Cloud Enterprise.

- Architected, prototyped, developed, Adobe Campaign on data platform using Java, Apache Tomcat, Spring Boot, MongoDB, CosmosDB, HDInside, Kafka, Postgres, Scala, Spark, Memos, Marathon, Docker, Kubernetes, New Relic, Splunk, microservices on Azure
- Deployed microservices on AWS through Adobe Ethos, AWS API Gateway, MongoDB, VPC peering

4/2016 – 10/2016, Cisco

Technical Leader

Cisco Streaming Integrator IoT platform to integrate applications and data streams so they can be bound together in workflows and processes to work in conjunction.

- Architected, prototyped, developed, Cisco streaming integrator platform using Java, Apache Tomcat, Servlets, JAX-RS, Kafka, Postgres, Mulesoft AnyPoint, Logstash, Elasticsearch, Mesos, Marathon, Docker.

3/2014 – 4/2016, Samsung Electronics America

Staff Engineer

Samsung SmartTV service builds TV profiles for Samsung SmartTV advertisement, sports notification, and recommendation.

- Architected, prototyped, developed, performance tuning data pipeline ingesting real-time data using Java, Python, Apache Tomcat, Servlets, JAX-RS, RabbitMQ, Apache Camel, Redis, Cassandra, AWS RDS Oracle, AWS Redshift on Amazon cloud platform.

4/2013 – 3/2014, Narus/Boeing/Symantec

Sr. Software Engineer

Narus cyber security big data analytics using Java, Servlets, JAX-RS, Hbase, Hadoop, Postgres on Cloudera platform.

- Developed data ingesting real-time server in Java, multi threading, Hbase, Hadoop, Postgres, thread dump analysis, performance tuning.
- Developed Narus portal in Java, Servlets, JAX-RS, Hbase, Hadoop, JSP, Html, JavaScript, Tomcat server.
- Set up performance benchmarks for Narus real-time server and tuned Hbase and Hadoop on Cloudera using Jmeter, Jprofiler, and Cloudera Admin console.

7/2010 – 3/2013, Ricoh

Sr. Software Engineer

Ricoh tablet backend web services using Java, Servlets, Soap, Zend PHP, MySQL, ActiveMQ, Camel, JDBC, Spring/Hibernate/JPA.

- Architected, prototyped, developed asynchronous web service calls for tablets to route documents using Java, ActiveMQ, Camel, Spring/Hibernate/JPA, Zend PHP5, MySQL.
- Architected, prototyped, developed document archive using Apache Camel S3 component/EIP.
- Architected, prototyped, developed Java Servlets, Tomcat to integrate with Sprint Mobile subscription SOAP web services.
- Architected, prototyped, developed Ricoh tablet 3G data usage for the alerting system and the billing system using Java, ActiveMQ, Camel, Zend PHP5.

2/2007 – 1/2010, VMware

Sr. Software Engineer

VMware Communities and VMworld Web 2.0 social networking software platform using Java, Struts 2.0, Servlet, Spring, JAX-WS 2, HTML, JavaScript, FreeMarker, MySQL, SQL Server, Ant, JUnit, IntelliJ.

- Developed single sign-on module on Communities platform to integrate with Oracle Fusion Middleware Oracle Access Manager authentication web services protocol and single sign-on module on VMworld platform to integrate with the 3rd party application.
- Developed server side business logic.
- Developed SOAP web services to integrate with VMware Virtual Appliance Marketplace in Drupal PHP framework using Struts 2 Java plugin module, Java, JAX-WS 2.0.
- Developed VMware product download applications for public evaluation using PHP 5, CakePHP, Akamai PHP APIs, SimpleTest PHP framework.

5/2005 – 1/2007, Yahoo, Inc.

Technical Yahoo

Yahoo analytic tools platform helps to know how visitors interact with Yahoo sites and to collect users big large data and to improve the site design and content through A/B testing.

- Developed client module embedded in Yahoo properties (Sports, HotJobs, Games, Mail, etc.) using Apache modules with C, C++, STL, CppUnit, XML, Perl, PHP 5, custom PHP extensions, Eclipse CDT.
- Developed server module using Java, Servlets, XML, Web Service REST, MySQL, Ant, JUnit, Eclipse on Tomcat.
- Developed reporting module using PHP 5, XML, XSLT, Oracle.

4/2003 – 4/2005, Yahoo, Inc.

PayDirect is an online person-to-person payment service.

- Developed PayDirect World Card, MoneyGram, Telecom account UI pages using PHP, HTML, JavaScript, LAMP.
- Developed PayDirect MoneyGram send money servlet using Java, JSP, Servlets, Apache Axis Web Services SOAP, XML, JDBC to download a fee table from the 3rd party MoneyGram fee server and load it into PayDirect Oracle database.

EDUCATION

Master of Computer Science, Syracuse University, Syracuse, New York

Bachelor of Computer Science, Soochow University, Taipei, Republic of China